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বিসিএস, ব্যাংক ও অন্যান্য সরকারি চাকুরির প্রস্তুতির

পূর্ণাঙ্গ লেকচার সিরিজ

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ইন্দ্রজিৎ চন্দ্র সরকার

৩৮তম বিসিএস

(সাধারণ শিক্ষা ক্যাডার)

প্রভাষক, গণিত বিভাগ

চিওড়া সরকারি কলেজ, কুমিল্লা।

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Video Class দেখুন



Study করুন



Live Exam দিন

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আজকের টপিক

লেকচার - ১০

টপিক:

Exponents and Logarithms

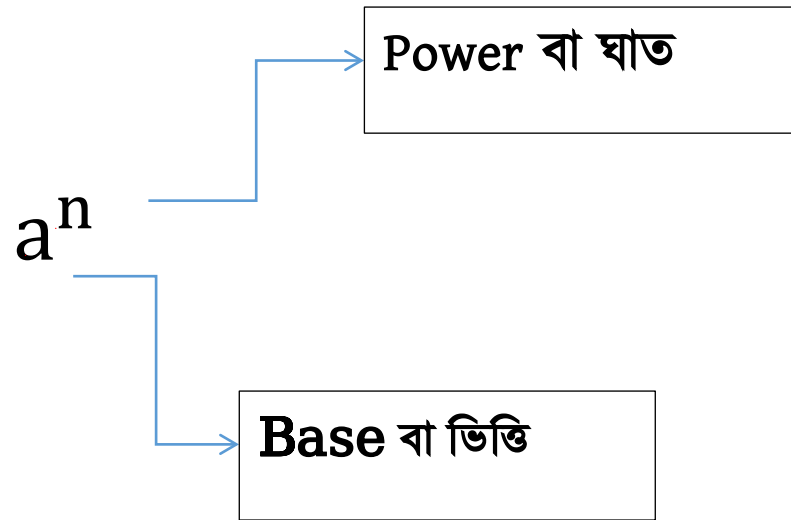
(সূচক ও লগারিদম)

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সূচক



Handwritten examples in red:
 $a^2 \rightarrow$ power
 $a \rightarrow$ base
 $a \rightarrow$ power
 $2 \rightarrow$ base

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বৈশিষ্ট্যঃ

$$\diamond \underline{a^m} \times \underline{a^n} = a^{m+n}, \quad a \neq 0$$

$$\diamond \frac{a^m}{a^n} = a^{m-n}$$

$$\diamond (a^n)^m = a^{mn}$$

$$\diamond \sqrt[n]{a} = a^{\frac{1}{n}}$$

$$\diamond (ab)^n = a^n b^n$$

$$\diamond \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$\diamond a^0 = 1, \quad a \neq 0$$

$$\diamond \text{যদি } a^x = b^x \text{ হয় তবে } a = b \text{ হবে।}$$

$$\diamond \text{যদি } \underline{a^x} = \underline{a^y} \text{ হয় তবে, } x = y \text{ হবে।}$$

$$\begin{aligned} a^2 \times a^3 &= a^{2+3} = a^5 \\ \frac{a^2}{a^3} &= a^{2-3} = a^{-1} \\ (a^2)^3 &= a^{2 \times 3} = a^6 \\ \sqrt[3]{a} &= a^{\frac{1}{3}} \end{aligned}$$

$$a^0 = 1$$

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০১। মান বের করুনঃ

$$\begin{aligned} &= \frac{5 \cdot 3^m - 3^2 \cdot 3^{m-1}}{3^m - 3^{m-1}} \\ &= \frac{5 \cdot 3^m - 3^{2+m-1}}{3^m - 3^m \cdot 3^{-1}} \\ &= \frac{5 \cdot 3^m - 3^{m+1}}{3^m (1 - 3^{-1})} \\ &= \frac{5 \cdot 3^m - 3^m \cdot 3^1}{3^m (1 - \frac{1}{3})} \end{aligned}$$

$$\frac{5 \cdot 3^m - 9 \cdot 3^{m-1}}{3^m - 3^{m-1}}$$

(base - same হতে হবে)

$$\left[\begin{array}{l} a^m \cdot a^n = a^{m+n} \\ a^{m+n} = a^m \cdot a^n \end{array} \right]$$

$$\begin{aligned} &= \frac{\cancel{3^m} (5 - 3)}{\cancel{3^m} \left(\frac{3-1}{3} \right)} \\ &= \frac{2}{\frac{2}{3}} \\ &= \cancel{2} \times \frac{3}{\cancel{2}} = 3 \end{aligned}$$

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নিজে করুন

$$\frac{3^{m+1}}{(3^m)^{m-1}} \div \frac{9^{m+1}}{(3^{m-1})^{m+1}}$$

$$= \frac{3^{m+1}}{3^{m^2-m}} \div \frac{(3^2)^{m+1}}{3^{m^2-1}}$$

$$= 3^{(m+1) - (m^2-m)} \div \frac{3^{2m+2}}{3^{m^2-1}}$$

$$= 3^{m+1-m^2+m} \div 3^{(2m+2) - (m^2-1)}$$

$$= 3^{2m+1-m^2} \div 3^{2m+2-m^2+1}$$

$$= 3^{2m+1-m^2} \div 3^{2m+3-m^2}$$

$$= 3^{2m+1-m^2 - (2m+3-m^2)}$$

$$= 3^{2m+1-m^2-2m-3+m^2}$$

$$= 3^{-2} = \frac{1}{3^2} = \frac{1}{9}$$



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Q21

$$\sqrt[n]{x} = x^{\frac{1}{n}}$$

$$bc \sqrt{\frac{\frac{b}{x\bar{c}}}{\frac{c}{x\bar{b}}}} \times ca \sqrt{\frac{\frac{c}{x\bar{a}}}{\frac{a}{x\bar{c}}}} \times ab \sqrt{\frac{\frac{a}{x\bar{b}}}{\frac{b}{x\bar{a}}}}$$

$$= bc \sqrt{x^{\frac{b}{c} - \frac{c}{b}}} \times ca \sqrt{x^{\frac{c}{a} - \frac{a}{c}}} \times ab \sqrt{x^{\frac{a}{b} - \frac{b}{a}}} \\ = \left(x^{\frac{b^2 - c^2}{bc}}\right)^{\frac{1}{bc}} \times \left(x^{\frac{c^2 - a^2}{ac}}\right)^{\frac{1}{ac}} \times \left(x^{\frac{a^2 - b^2}{ab}}\right)^{\frac{1}{ab}}$$

$$= x^{\frac{b^2 - c^2}{b^2 c^2}} \times x^{\frac{c^2 - a^2}{a^2 c^2}} \times x^{\frac{a^2 - b^2}{a^2 b^2}}$$

$$= x^{\frac{a^2(b^2 - c^2) + b^2(c^2 - a^2) + c^2(a^2 - b^2)}{a^2 b^2 c^2}}$$

$$= x^{\frac{0}{a^2 b^2 c^2}}$$

$$= x^0 = 1$$

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৩। $a^x = b$, $b^y = c$ এবং $c^z = a$ হলে, xyz এর মান বের করুন।

Solⁿ চাওয়া আছে,

$$a^x = b$$

$$\text{বা, } (a^x)^y = b^y$$

$$\text{অ, } a^{xy} = c \quad [\because b^y = c]$$

$$\text{বা, } (a^{xy})^z = c^z$$

$$\text{অ, } \underline{a^{xyz}} = \underline{a^1} \quad [\because c^z = a]$$

$$\text{অ, } xyz = 1 \quad [\because a^n = a^m \text{ তাহলে } n = m] =$$

$$\therefore \underline{\underline{\text{নির্ণেয় মান} = 1}}$$

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নিজে করুন

~~x~~ $x^a = y^b = z^c$ এবং $xyz = 1$ হলে, $ab + bc + ca$ এর মান বের করুন।

ধি, $x^a = y^b = z^c = k$

অতঃ, $x^a = k$

অ, $(x^a)^{\frac{1}{a}} = k^{\frac{1}{a}}$

অ, $x^{a \times \frac{1}{a}} = k^{\frac{1}{a}}$

$\therefore x = k^{\frac{1}{a}}$

অতঃ, $y^b = k$

অ, $(y^b)^{\frac{1}{b}} = k^{\frac{1}{b}}$

$\therefore y = k^{\frac{1}{b}}$

অতঃ, $z = k^{\frac{1}{c}}$

এখন,
 $xyz = k^{\frac{1}{a}} \cdot k^{\frac{1}{b}} \cdot k^{\frac{1}{c}}$

অ, $1 = k^{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$

অ, $k^0 = k$

অ, $0 = \frac{bc + ac + ab}{abc}$

$\therefore ab + bc + ca = 0 \times abc$
 $= 0$

Ans. 0.

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8। $p = xy^{a-1}, q = xy^{b-1}, r = xy^{c-1}$ হলে, $\left(\frac{p}{q}\right)^c \times \left(\frac{q}{r}\right)^a \times \left(\frac{r}{p}\right)^b = ?$

$$\begin{aligned} \text{L.H.S} &= \left(\frac{p}{q}\right)^c \times \left(\frac{q}{r}\right)^a \times \left(\frac{r}{p}\right)^b \\ &= \left(\frac{xy^{a-1}}{xy^{b-1}}\right)^c \times \left(\frac{xy^{b-1}}{xy^{c-1}}\right)^a \times \left(\frac{xy^{c-1}}{xy^{a-1}}\right)^b \\ &= \left\{y^{(a-1)-(b-1)}\right\}^c \times \left\{y^{b-1-(c-1)}\right\}^a \times \left\{y^{c-1-(a-1)}\right\}^b \\ &= \left(y^{a-1-b+1}\right)^c \times \left(y^{b-1-c+1}\right)^a \times \left(y^{c-1-a+1}\right)^b \\ &= \left(y^{a-b}\right)^c \times \left(y^{b-c}\right)^a \times \left(y^{c-a}\right)^b \\ &= y^{ac-bc} \times y^{ba-ca} \times y^{bc-ab} \\ &= y^{ac-bc+ab-ca+bc-ab} \\ &= y^0 = \underline{\underline{1 \text{ Ans}}}} \end{aligned}$$



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নিজে করুন

~~১~~ $a^{\frac{1}{x}} = b^{\frac{1}{y}} = c^{\frac{1}{z}}$ এবং $abc = 1$ হয়, তবে দেখাও যে, $x + y + z = 0$

solⁿ িতি, $a^{\frac{1}{x}} = b^{\frac{1}{y}} = c^{\frac{1}{z}} = k$

$$a^{\frac{1}{x}} = k$$

$$\therefore, \left(a^{\frac{1}{x}}\right)^x = k^x$$

$$\therefore a = k^x$$

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৫। সমাধান করঃ $4^x - 3(2^{x+2}) + 2^5 = 0$

Solⁿ $4^x - 3(2^{x+2}) + 2^5 = 0$

৩, $(2^2)^x - 3 \cdot 2^x \cdot 2^2 + 32 = 0$

৩, $(2^x)^2 - 3 \cdot 2^x \cdot 4 + 32 = 0$

৩, $(2^x)^2 - 12 \cdot 2^x + 32 = 0$

ধরি, $2^x = y$ তাহলে,

$$y^2 - 12y + 32 = 0$$

$$৩, y^2 - 8y - 4y + 32 = 0$$

$$৩, y(y-8) - 4(y-8) = 0$$

$$\therefore (y-8)(y-4) = 0$$

base - ওল্ড ২১০ ২১৮

২১, $y - 8 = 0$

৩, $y = 8$

৩, $2^x = 2^3$

$\therefore x = 3$

অথবা,

$$y - 4 = 0$$

৩, $y = 4$

৩, $2^x = 2^2$

$\therefore x = 2$

$\therefore$ নির্দিষ্ট সমাধান,

$x = 2, 3$ Ans.



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নিজে করুন

$$\begin{aligned} & \cancel{\sqrt{3}}^{x+1} = (\sqrt[3]{3})^{2x-1} \\ \text{or } & \left(3^{\frac{1}{2}}\right)^{x+1} = \left(3^{\frac{1}{3}}\right)^{2x-1} \\ \text{or, } & 3^{\frac{x+1}{2}} = 3^{\frac{2x-1}{3}} \\ \therefore & \frac{x+1}{2} = \frac{2x-1}{3} \end{aligned}$$



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লগারিদম

$a^x = N$, ($a > 0$, $a \neq 1$) হলে, $x = \log_a N$ কে N এর a ভিত্তিক লগ বলা হয়।

দ্রষ্টব্যঃ x ধনাত্মক বা ঋণাত্মক যাই হোক না কেন, $a > 0$ হলে সর্বদা a^x ধনাত্মক। তাই শুধু ধনাত্মক সংখ্যারই লগের মান আছে যা বাস্তব। শূন্য বা ঋণাত্মক সংখ্যার লগের বাস্তব মান নেই।

$$\frac{2 + -x \leq \text{syn error}}{0} \rightarrow \text{Math error}$$

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বৈশিষ্ট্য

➤ $\log_a M + \log_a N = \log_a (MN)$

➤ $\log_a M - \log_a N = \log_a \left(\frac{M}{N}\right)$

➤ $\log_a M^n = n \log_a M \Rightarrow$

➤ $\log_a 1 = 0 \Rightarrow$

➤ $\log_a a^n = n$

➤ $\log_a Y = x$ হলে, $a^x = y$

➤ $\log_a b = \frac{1}{\log_b a}$

➤ বিঃদ্রঃ ($\log$ এর বেস বা ভিত্তি e হলে তা $\ln$ হবে) অর্থাৎ $\log_e X = \ln X$

$\log_{10} a^2 = 2 \log_{10} a$

$\log_a 1 = \log_a a^0 = 0 \cdot \log_a a = 0 \cdot 1 = 0$

$\log_2 8 = 3 \therefore 8 = 2^3$

$\log_a b \times \log_b a = \log_a a = 1$

$\log_e X = \ln X$



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১। $\log_{2\sqrt{5}} 400$ এর মান কত?

$$\begin{aligned} &= \log_{2\sqrt{5}} (20)^2 \\ &= \log_{2\sqrt{5}} (4 \times 5)^2 \\ &= \log_{2\sqrt{5}} \{(2\sqrt{5})^2\}^2 \\ &= \log_{2\sqrt{5}} (2\sqrt{5})^4 \\ &= 4 \times 1 \\ &= 4 \end{aligned}$$



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নিজে করুন

 $\log_5 \sqrt[3]{5}, \log_{3\sqrt{2}} 324, \log_3 \frac{1}{9}$

$$\log_5 \sqrt[3]{5}$$

$$= \log_5 5^{\frac{1}{3}}$$

$$= \frac{1}{3} \log_5 5$$

$$= \frac{1}{3} \times 1$$

$$= \frac{1}{3}$$

$$\log_{3\sqrt{2}} 324$$

$$= ?$$



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সমাধান করুন

$$২। \log_{10} x = -3$$

$$\text{ক, } x = 10^{-3}$$

$$\therefore x = \frac{1}{10^3}$$

$$= \frac{1}{1000}$$

$$\left[\log_a x = n \text{ হলে } x = a^n \right]$$



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নিজে করুন

$$\log_x 324 = 4, \log_{\sqrt{8}} x = 3\frac{1}{3},$$

≡

$$\log_{\sqrt{8}} x = 3\frac{1}{3} = \frac{10}{3}$$

$$\begin{aligned} \therefore, x &= (\sqrt{8})^{\frac{10}{3}} \\ &= \left(8^{\frac{1}{2}}\right)^{\frac{10}{3}} \\ &= 8^{\frac{10}{2 \times 3}} \\ &= (2^3)^{\frac{5}{3}} \\ &= 2^{3 \times \frac{5}{3}} \\ &= 2^5 \\ &= 32 \end{aligned}$$

Ans

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$$৩। \frac{\log_{10} \sqrt{27} + \log_{10} 8 - \log_{10} \sqrt{1000}}{\log_{10} 1.2}$$

$$= \frac{\log_{10} (27)^{\frac{1}{2}} + \log_{10} 2^3 - \log_{10} (1000)^{\frac{1}{2}}}{\log_{10} \frac{126}{105}}$$

$$= \frac{\log_{10} (3^3)^{\frac{1}{2}} + 3 \log_{10} 2 - \log_{10} (10^3)^{\frac{1}{2}}}{\log_{10} 6 - \log_{10} 5}$$

$$= \frac{\log_{10} 3^{\frac{3}{2}} + 3 \log_{10} 2 - \frac{3}{2} \log_{10} (2 \times 5)}{\log_{10} (3 \times 2) - \log_{10} 5}$$

$$= \frac{\frac{3}{2} \log_{10} 3 + 3 \log_{10} 2 - \frac{3}{2} \log_{10} 2 - \frac{3}{2} \log_{10} 5}{\log_{10} 3 + \log_{10} 2 - \log_{10} 5}$$

$$= \frac{\frac{3}{2} \log_{10} 3 + (3 - \frac{3}{2}) \log_{10} 2 - \frac{3}{2} \log_{10} 5}{\log_{10} 3 + \log_{10} 2 - \log_{10} 5}$$

$$= \frac{\frac{3}{2} (\log_{10} 3 + \log_{10} 2 - \log_{10} 5)}{\log_{10} 3 + \log_{10} 2 - \log_{10} 5}$$

$$= \frac{3}{2}$$

$$= \frac{3}{2}$$

$$= \frac{3}{2}$$

Ans

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নিজে করুন

$$\log_{\sqrt{a}} b \times \log_{\sqrt{b}} c \times \log_{\sqrt{c}} a = ? //$$

$$= \log_{\sqrt{a}} b \times \log_{\sqrt{b}} c \times \log_{\sqrt{c}} a$$

$$= \log_{\sqrt{a}} (\sqrt{b})^2 \times \log_{\sqrt{b}} (\sqrt{c})^2 \times \log_{\sqrt{c}} (\sqrt{a})^2$$

$$= 2 \log_{\sqrt{a}} \sqrt{b} \times 2 \log_{\sqrt{b}} \sqrt{c} \times 2 \log_{\sqrt{c}} \sqrt{a}$$

$$= 8 \times \log_{\sqrt{a}} \sqrt{b} \times \log_{\sqrt{b}} \sqrt{c} \times \log_{\sqrt{c}} \sqrt{a}$$

$$= 8 \times \log_{\sqrt{a}} \sqrt{c} \times \log_{\sqrt{c}} \sqrt{a}$$

$$= 8 \times \log_{\sqrt{a}} \sqrt{a}$$

$$= 8 \times 1 = 8$$

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8। $p = xy^{a-1}, q = xy^{b-1}, r = xy^{c-1}$ হলে প্রমাণ করুন, $\log p^{b-c} + \log q^{c-a} + \log r^{a-b} = 0$

$$\begin{aligned}
 \text{L.H.S} &= \log p^{b-c} + \log q^{c-a} + \log r^{a-b} \\
 &= \log (xy^{a-1})^{b-c} + \log (xy^{b-1})^{c-a} + \log (xy^{c-1})^{a-b} \\
 &= (b-c) \log xy^{a-1} + (c-a) \log xy^{b-1} + (a-b) \log xy^{c-1} \\
 &= (b-c) (\log x + \log y^{a-1}) + (c-a) (\log x + \log y^{b-1}) \\
 &\quad + (a-b) (\log x + \log y^{c-1}) \\
 &= (b-c) \log x + (b-c)(a-1) \log y + (c-a) \log x + (c-a)(b-1) \log y \\
 &\quad + (a-b) \log x + (a-b)(c-1) \log y \\
 &= (b-c + c-a + a-b) \log x + (a-b-b-c-a + c + b-c - c - a + a + a - a - b + b) \log y \\
 &= 0 \cdot \log x + 0 \log y = 0
 \end{aligned}$$



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নিজে করুন

যদি $x = 1 + \log_a(bc)$, $y = 1 + \log_b(ca)$, $z = 1 + \log_c(ab)$ হয়, তবে প্রমাণ করুন
যে, $xyz = xy + yz + zx$

Solⁿ

$$\begin{aligned}x &= 1 + \log_a(bc) \\&= \log_a a + \log_a(bc) \\&= \log_a abc\end{aligned}$$

[$\log MN = \log M + \log N$]

$$\begin{aligned}\therefore \log_a abc &= x \\ \therefore abc &= a^x\end{aligned}$$
$$\begin{aligned}abc &= b^y \\ abc &= c^z\end{aligned}$$
$$a^x = b^y = c^z$$



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৫। যদি, $\frac{\ln a}{q-r} = \frac{\ln b}{r-p} = \frac{\ln c}{p-q}$ হলে, প্রমাণ করুন যে, $\underline{a^{q+r}} \cdot \underline{b^{r+p}} \cdot c^{p+q} = 1$

$$\text{ধরি, } \frac{\ln a}{q-r} = \frac{\ln b}{r-p} = \frac{\ln c}{p-q} = k$$

$$\therefore \frac{\ln a}{q-r} = k$$

$$\therefore \ln a = k(q-r)$$

$$\therefore (q+r) \ln a = k(q-r)(q+r)$$

$$\therefore \ln a^{q+r} = k(q^2 - r^2) \quad \text{--- (1)}$$

$$\text{অনুরূপ, } \frac{\ln b}{r-p} = k$$

$$\therefore \ln b = k(r-p)$$

$$\therefore (r+p) \ln b = k(r-p)(r+p)$$

$$\therefore \ln b^{r+p} = k(r^2 - p^2) \quad \text{--- (2)}$$

$$\text{অনুরূপভাবে, } \ln c^{p+q} = k(p^2 - q^2) \quad \text{--- (3)}$$

$$\begin{aligned} & \ln a^{q+r} + \ln b^{r+p} \\ & + \ln c^{p+q} = \\ & k(q^2 - r^2) + k(r^2 - p^2) \\ & + k(p^2 - q^2) \\ & \therefore \ln a^{q+r} \cdot b^{r+p} \cdot c^{p+q} \\ & = 0 = \ln 1 \end{aligned}$$

$$\therefore a^{q+r} \cdot b^{r+p} \cdot c^{p+q} = 1$$

[প্রমাণিত]



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নিজে করুন,

$\frac{\log_k a}{b-c} = \frac{\log_k b}{c-a} = \frac{\log_k c}{a-b}$ হলে, প্রমাণ করুন, $\underline{\underline{a^a \cdot b^b \cdot c^c = 1}}$

$$\frac{\log_k a}{b-c} = P$$

$$\log_k a = P(b-c)$$

$$\therefore a \log_k a = aP(b-c)$$

$$\therefore \log_k a^a = aP(b-c) \text{ --- (1)}$$

$$\log_k 1$$



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ধন্যবাদ

ঘৰে বসে বিসিএস ও অন্যান্য চাকৰিৰ লিখিত প্ৰস্তুতি নিতে ▶



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